

f) $\{\{\{2\}\}\}$ $\{\{2\}\}$ and 2 are not the same. $\therefore$ No

11) Determine whether each of these statements is true or false.

- a) $0 \in \emptyset$. By defn, a null set ($\emptyset$) has no elements. $\therefore$ The statement is False.
 b) $\emptyset \in \{\emptyset\}$. $\emptyset$ is the only member of $\{\emptyset\}$. $\therefore$ The statement is False.
 c) $\{\emptyset\} \subset \emptyset$. $\emptyset$ has no elements, by defn. $\therefore$ Its only subset is $\emptyset$. $\therefore$ The statement is False, by defn. of proper subset.
 d) $\emptyset \subset \{\emptyset\}$. The subsets of set $\{\emptyset\}$ is $\emptyset$ and $\{\emptyset\}$. $\emptyset$ and $\{\emptyset\}$ are not same. $\therefore$ The statement is True, by defn. of proper subset.
 e) $\{\emptyset\} \in \{\emptyset\}$. The set $\{\emptyset\}$ has no such member $\{\emptyset\}$. $\therefore$ The statement is False.
 f) $\{\emptyset\} \subset \{\emptyset\}$. Both the sets are same. $\therefore$ The statement is False by defn. of proper subset.
 g) $\{\emptyset\} \subseteq \{\emptyset\}$. By defn. of subset, and Theorem 1, The statement is True.

12) Determine whether these statements are true or false.

- a) $\emptyset \in \{\emptyset\}$. $\emptyset$ is a member of $\{\emptyset\}$. $\therefore$ The statement is a True one.
 b) $\emptyset \in \{\emptyset, \{\emptyset\}\}$. $\emptyset$ is a member of $\{\emptyset, \{\emptyset\}\}$. $\therefore$ The statement is True.
 c) $\{\emptyset\} \in \{\emptyset\}$. $\{\emptyset\}$ has only $\emptyset$ as a member, not $\{\emptyset\}$. $\therefore$ The statement is False.
 d) $\{\emptyset\} \in \{\{\emptyset\}\}$. It's easy to see that this statement is True, because $\{\emptyset\}$ is the only member of the set $\{\{\emptyset\}\}$.
 e) $\{\emptyset\} \subset \{\emptyset, \{\emptyset\}\}$. The members of the set $\{\emptyset, \{\emptyset\}\}$ are $\emptyset$ and $\{\emptyset\}$. The only member of $\{\emptyset\}$ is $\emptyset$. $\therefore$ By Definition 3 and from the definition of the proper subset, the given statement is True.
 f) $\{\{\emptyset\}\} \subset \{\emptyset, \{\emptyset\}\}$. The only member of the set $\{\{\emptyset\}\}$ is $\{\emptyset\}$. The members of the set $\{\emptyset, \{\emptyset\}\}$ are $\emptyset$ and $\{\emptyset\}$. $\therefore$ By Definition 3 and from the definition of the proper subset, the given statement is True.
 g) $\{\{\emptyset\}\} \subset \{\{\emptyset\}, \{\emptyset\}\}$. The only member of both the given sets is $\{\emptyset\}$. $\therefore$ By Definition 3 and from the definition.